

Introduction to the Internet and Email Basics

1. Introduction to the Internet

What is the Internet?

- A global network of interconnected computers that communicate via standardized protocols.
- Originated from **ARPANET** in the 1960s.
- Facilitates communication, file sharing, web browsing, and business transactions.
- A **global network** that connects millions of computers for **communication and data sharing**.
- Uses standardized protocols such as **TCP/IP** for transmitting data.

Key Terms:

Basic Internet Terminology

- **Peer to Peer Network:** A network connecting two devices.
- **LAN (Local Area Network):** A network connecting computers in a small area (e.g., an office).
- **WAN (Wide Area Network):** A network that spans a large geographical area (e.g., the internet).
- **Servers & Clients:**
 - **Server:** A computer that stores and provides resources (websites, emails, files).
 - **Client:** A device (computer, phone) that requests services from a server.
- **IP Addresses & Domains:**
 - An **IP address** (e.g., 192.168.1.1) is a unique identifier for devices on a network.
- **Internet Service Providers (ISP):** Companies that provide internet access (e.g., Safaricom, AT&T, Comcast).

- **Bandwidth & Speed:**

- **Bandwidth:** The amount of data that can be transmitted per second (measured in Mbps/Gbps).
 - Higher bandwidth allows faster data transfer.
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2. Basics of Internet Protocols

What Are Protocols?

- Rules and standards that allow communication between devices.
 - **Types of Internet Protocols:**
 - **TCP/IP (Transmission Control Protocol/Internet Protocol):** The foundation of internet communication.
 - **HTTP (Hypertext Transfer Protocol):** Used for web browsing (e.g., accessing web pages).
 - **HTTPS (Secure HTTP):** Encrypted version of HTTP for secure communication (e.g., online banking).
 - **FTP (File Transfer Protocol):** Transfers files between computers.
 - **DNS (Domain Name System):** Converts domain names to IP addresses (e.g., google.com → 142.250.180.206).
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3. Web Browsers & Search Engines

- **WWW (World Wide Web):** A system of interlinked web pages accessible via the internet.
- **HTTP (Hypertext Transfer Protocol):** A protocol used to access web pages.
- **HTML (Hypertext Markup Language):** The coding language used to structure web pages.
- **Webpage:** A digital document that is linked to the World Wide Web and viewable by anyone connected to the internet having a web browser.
- **Website:** A collection of related web pages stored on a web server (e.g., www.google.com).
- **Surfing:** The act of browsing different web pages on the internet.

Web Browsers

- Software applications used to access the internet (e.g., Chrome, Firefox, Edge, Safari).
- **Components of a Web Browser:**
 - **Address Bar:** Where URLs are entered.
 - **Bookmarks:** Save frequently visited pages.
 - **Cookies:** Small data stored by websites for personalization.
- **Incognito/Private Mode:** Browsing without storing history or cookies.

Browser Functions

- **Stop Button:** Stops loading a web page.
- **Refresh Button:** Reloads a web page to show updated content.

Search Engines & Effective Searching

- **Definition:** Programs that allow users to find information on the internet.
 - **Examples:** Google, Bing, Yahoo.
 - **Search Tips:**
 - Use **quotation marks ("")** for exact phrases.
 - Use **Boolean operators:**
 - **AND** (e.g., "internet AND email") – finds results with both terms.
 - **OR** (e.g., "internet OR web") – finds results with either term.
 - **NOT** (e.g., "email NOT spam") – excludes a term.
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4. Email Basics

What is Email?

- A digital communication tool that allows users to send and receive messages over the internet.
- Faster and more reliable than traditional mail.

Structure of an Email Address

- **Format:** username@domain.com
 - username – Identifies the user.
 - domain.com – Identifies the email service provider (e.g., Gmail, Yahoo, Outlook).

Valid vs. Invalid Email and Web Addresses

- **Valid:**
 - stella@yahoo.com
 - www.hotelskenya.co.ke
 - www.mtkenya.org
 - www.safaricom.co.ke
- **Invalid:**
 - www.@gmail.com (Incorrect placement of @)
 - eric.@www.gmail@yahoo.com (Too many special characters)
 - a@b@c@gmail.com (Multiple @ symbols)

Email Clients & Services

- **Web-Based Clients:** Gmail, Yahoo Mail, Outlook.com.
- **Desktop Clients:** Microsoft Outlook, Mozilla Thunderbird.

Email Protocols

- **SMTP (Simple Mail Transfer Protocol):** Used to send emails.
- **POP3 (Post Office Protocol v3):** Downloads emails from the server to a device.
- **IMAP (Internet Message Access Protocol):** Synchronizes emails across multiple devices.

Email Components

- **To:** Main recipient(s).
- **CC (Carbon Copy):** Sends a copy to additional recipients.
- **BCC (Blind Carbon Copy):** Sends a copy without revealing recipients.
- **Subject:** The main topic of the email.

- **Attachments:** Files (documents, images, etc.) sent with the email.
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5. Uses of the Internet

What You Can Do with the Internet

1. **Communication:** Emails, messaging apps, video calls.
2. **Research & Information:** Access to online books, articles, and educational resources.
3. **E-Commerce:** Buying and selling goods online.
4. **Entertainment:** Streaming music, videos, and online gaming.
5. **Banking & Financial Transactions:** Online banking and bill payments.

Advantages & Disadvantages of Internet Access at Home

Advantages:

1. **Convenience:** Access to information and services from home.
2. **Communication:** Easy to stay connected via email, social media, and video calls.

Disadvantages:

1. **Security Risks:** Exposure to cyber threats such as hacking and phishing.
2. **Distractions:** Excessive use may lead to decreased productivity.

5. Cybersecurity & Safe Internet Use

Common Internet Threats

1. **Phishing Attacks:**
 - Fake emails or websites that steal login credentials.
 - **Prevention:** Check sender details, avoid suspicious links.
2. **Malware & Viruses:**
 - Harmful software that can damage data or steal information.
 - **Prevention:** Use antivirus software, avoid unknown downloads.
3. **Spam Emails:**
 - Unwanted or fraudulent emails, often promoting scams.

- **Prevention:** Use spam filters and avoid clicking unknown links.

4. Identity Theft:

- Cybercriminals steal personal information for fraud.
- **Prevention:** Enable **two-factor authentication (2FA)** and use strong passwords.

Best Practices for Safe Internet Use

- **Use Strong Passwords:**
 - Mix uppercase, lowercase, numbers, and symbols.
 - Avoid common words (e.g., "password123").
 - **Enable 2FA:** Adds an extra layer of security for logins.
 - **Recognize Secure Websites:** Look for **HTTPS** and a padlock icon.
 - **Keep Software & Browsers Updated:** Prevents security vulnerabilities.
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Lecture Wrap-Up

- Encourage students to explore **free learning resources:**
 - **Mozilla Web Literacy Guide** (<https://learning.mozilla.org/>)
 - **Google's Internet Safety Tips** (<https://safety.google/>)
 - **Email Security Training by Microsoft** (<https://support.microsoft.com/>)
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